Adventures in 8K

Prepared for syd<video>

Microsoft Reactor April 2024

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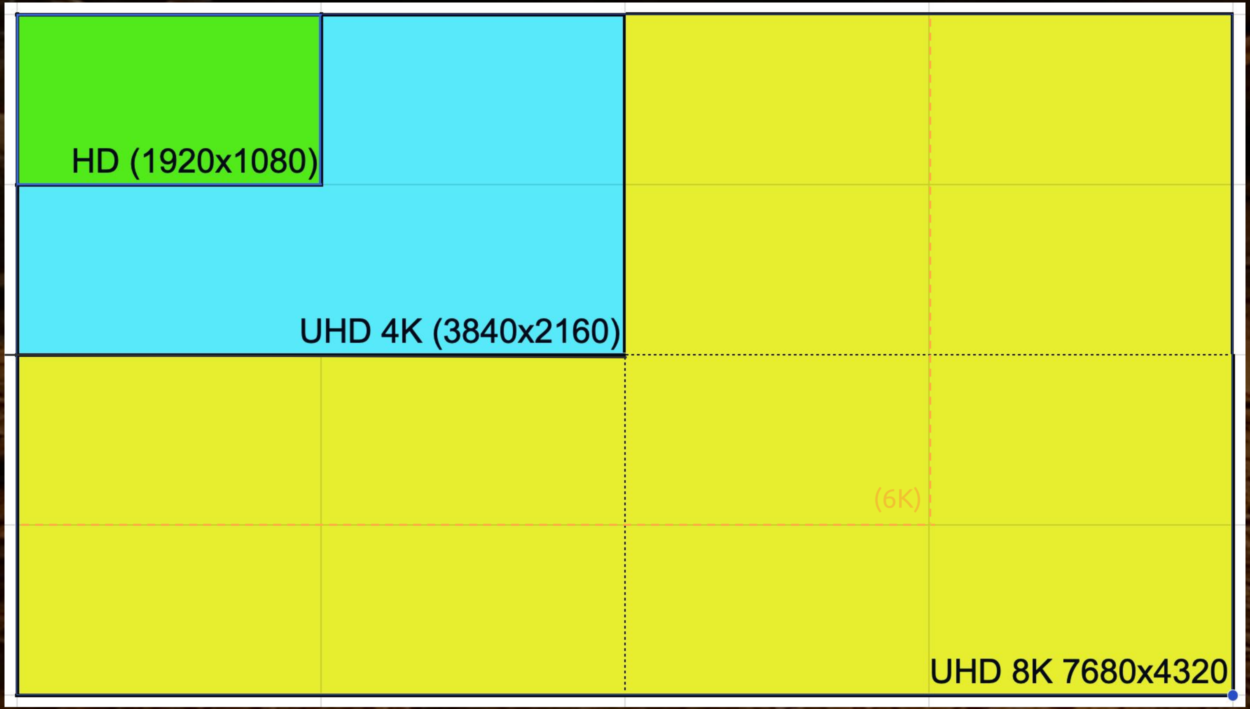
What's **8K?** A bit like a stack of *sixteen* 1080HD TVs!

8K UHD-2
(4320p)

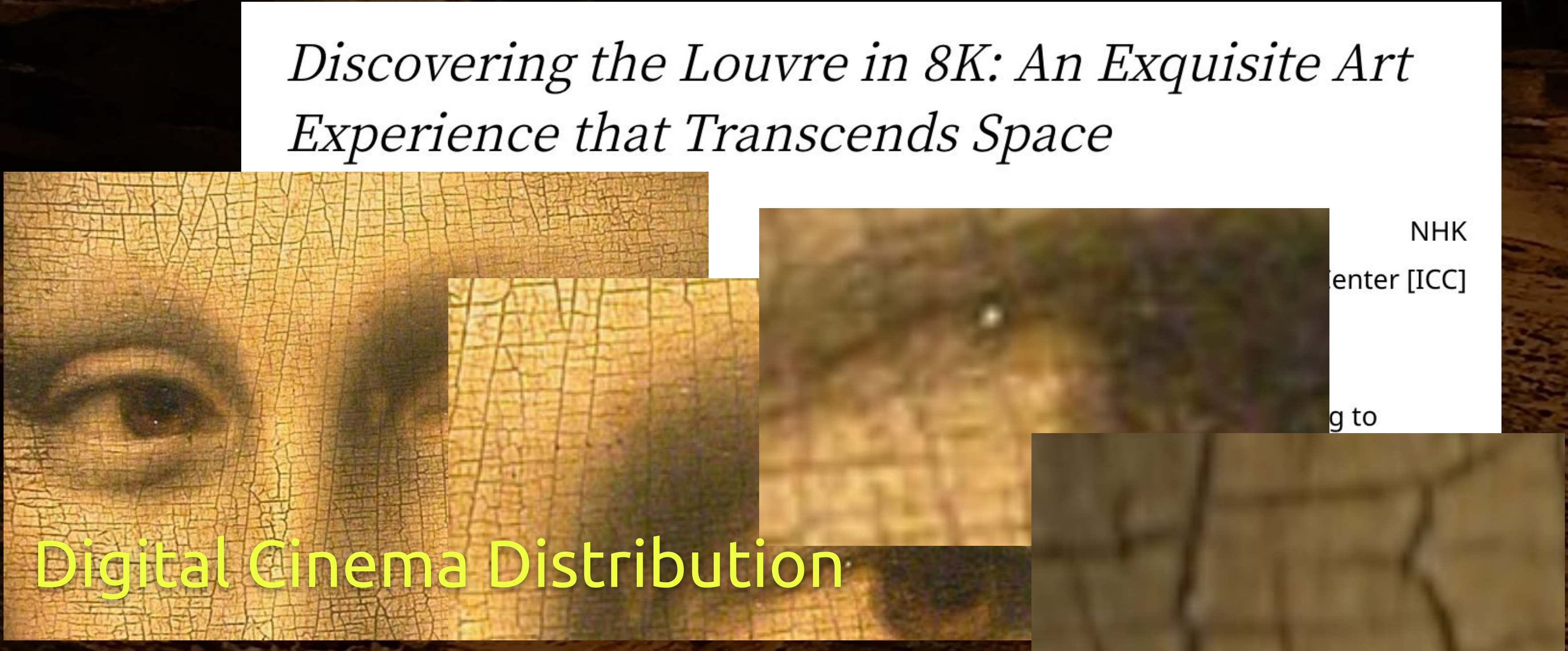
= 4 x 4K
(2160p)

= 16 x HD
(1080p)



8K Why?! Looks absolutely fabulous! *(no, really)*

Early demos at IBC - *actually* **overwhelming**



8K Why?! So, does anyone even *want* 8K?

Early demos at IBC - *actually* overwhelming

Seems 8K is really great for art! (amazeballs)
Eye bursting 8K home theatre & bijou cinema!

- but, **do we all need 8K TVs at home?**

No! we do not *need* 8K TVs! - well captured 4K at good bitrate is enough pixel density for 2D

8K Why *else*... Broadcast channel sharing (**mosaic**)

Better overall **aggregate compression**
Lower satellite/terrestrial bandwidth costs
and storage, by combining multiple encodings

Encrypted transport can be a grid of four x 4K services or maybe a pick and mix of say two x 4K and eight x HD services *oblivious to the end viewers*

Players crop and switch the audio tracks

8K Why ***really***... Proper immersive **spatial video**

VR is panning/zooming in on only a *small section* of a larger visual field, it needs hi-res

1080 VR180° is *crap* - 4K *isn't that great either*

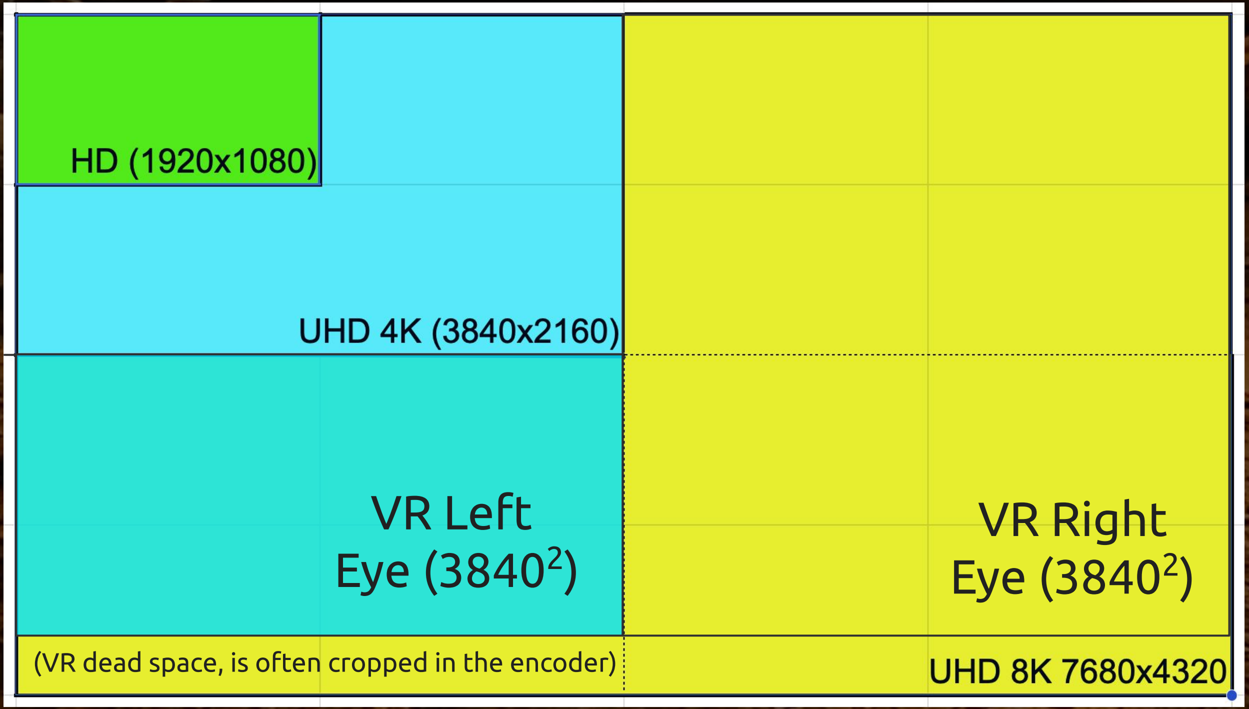
Higher PPD is vital when each eye has its own $3k^2$ px microOLED

What's **8K Spatial VR?** Side-by-side square 4K TVs*

8K UHD-2 (4320p)

= **2** x **4K** Sq (3840p)

* can be top/below but side-by-side is most compatible



Big problem today is there's not much 8K content

- What are the problems with 8K
- Solutions we have today (various budgets)
- What's the future of 8K looking like?

8K - Some challenges

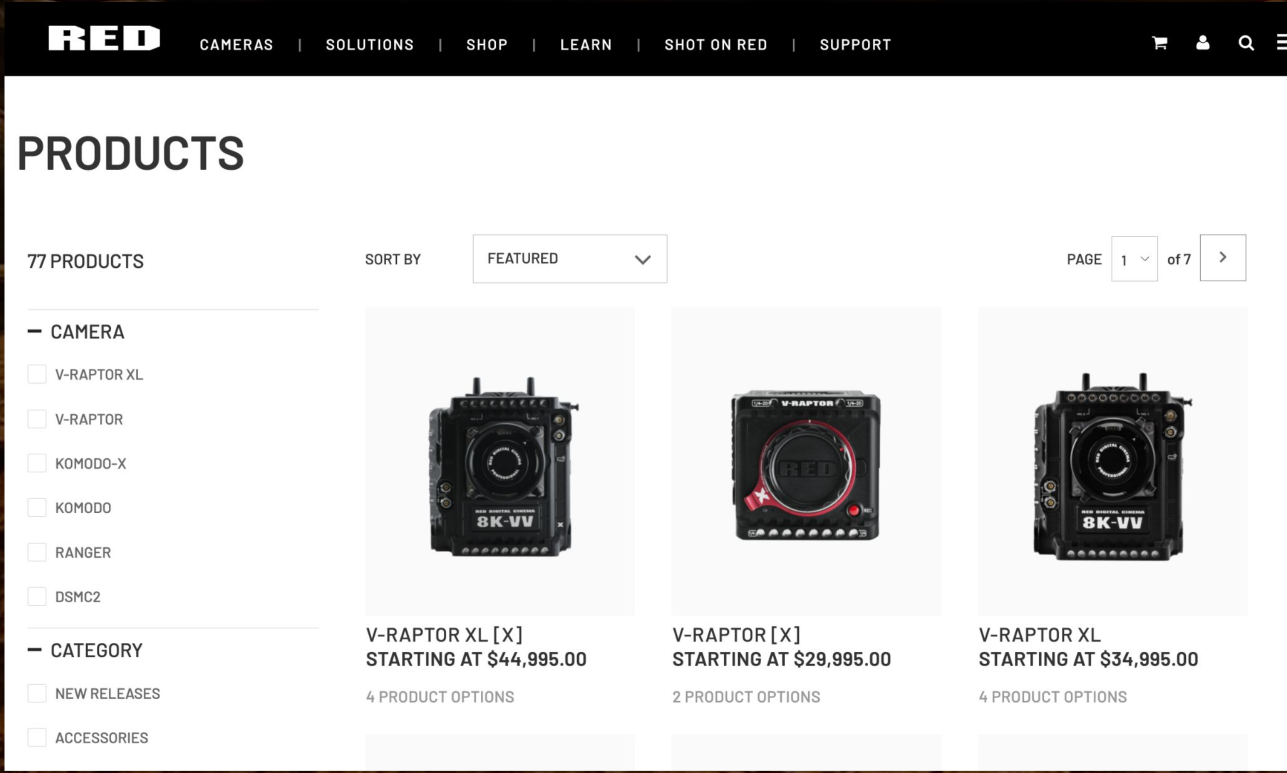
Higher upfront and operational expenses - capturing and pushing **7680** x **4320** (33.2 Megapixels) around the planet ain't cheap

RAW bitrate (12-bit) @ 60 fps is 3.89 Gbps

You're gonna need a good hardware codec! And plenty storage/bandwidth

8K cons - We're talking significant dollarydoos $

e.g. latest 8K camera AU$**65,000** RAPTOR XL (RED/Nikon)



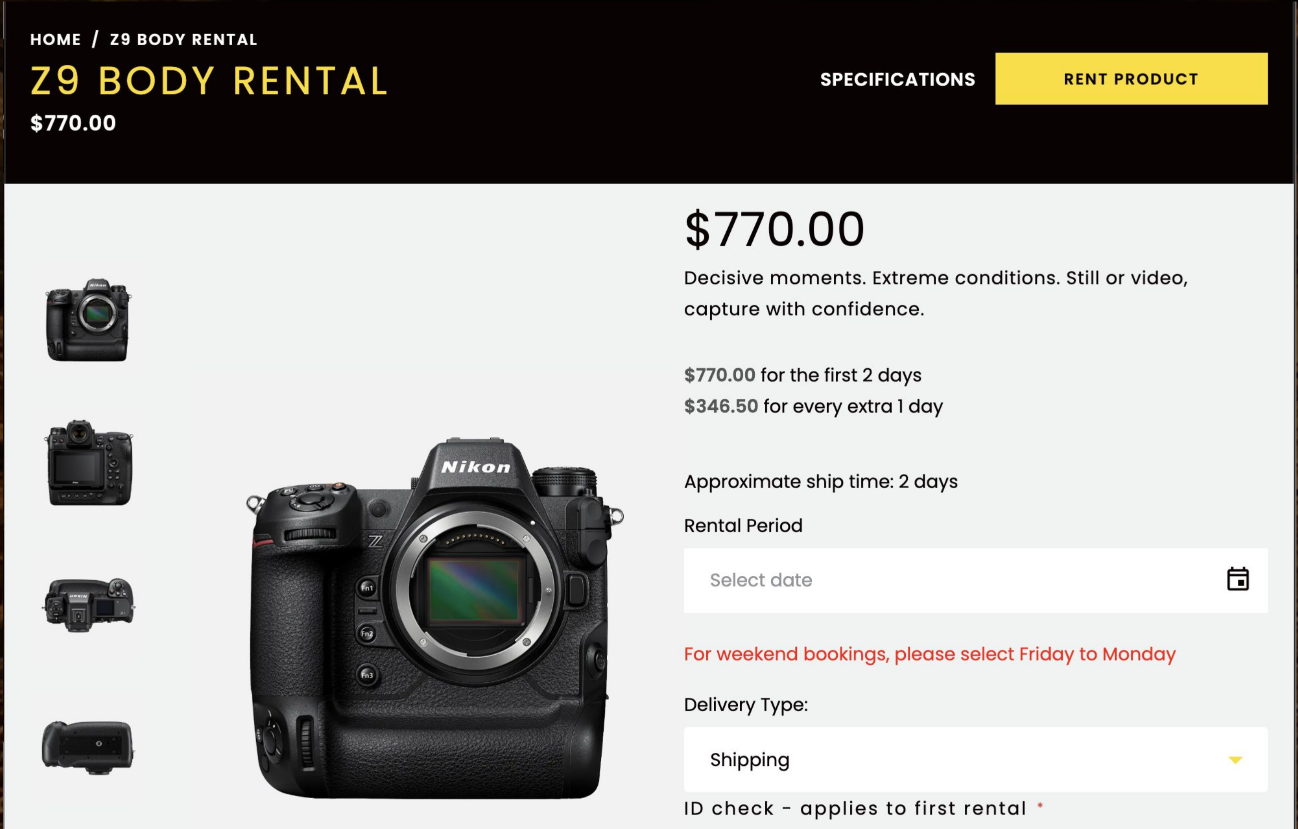
George's Sydney had a slightly older V-RAPTOR AU$42,000

8K cons - Even renting gear adds up quick

e.g. rental AU$**385** per day

> AU$**1,500**/wk

(Nikon Australia)



8K cons - need a lens for your camera body too

buy

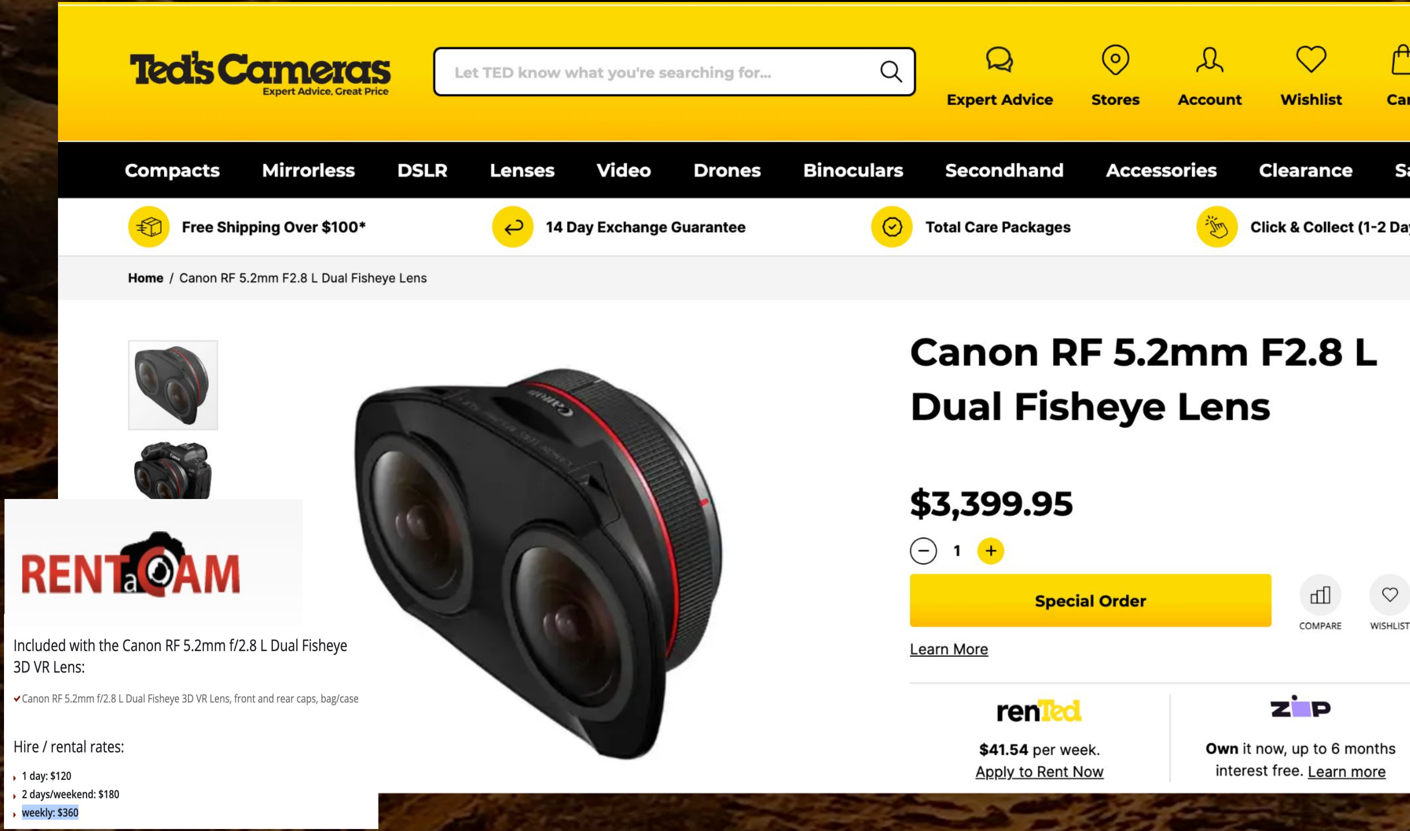
AU$**3,400**

rent

AU$**360**/wk

(per camera)

This all gets heavy too!



8K cons - needs plenty fast digital storage

Wanna edit? storage has come a long way, but still a significant outlay

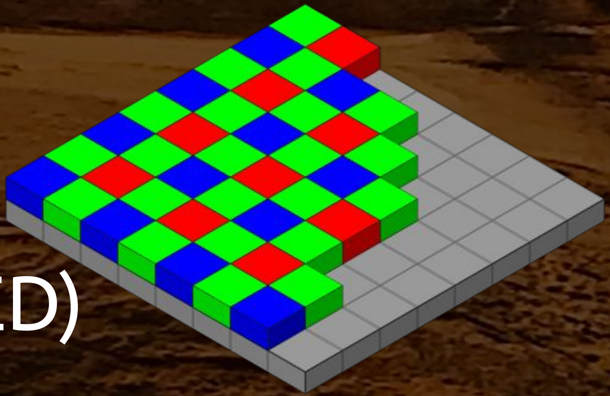
AU$**750** (4TB 2Gb/s SSD)

(these run HOT too, battery thirsty)

8K cons - Patent encumbered *(just be aware)*

De-mosaicing algorithms, still needed when dealing with larger bayer sensor datasets

e.g. methods to derive green from only red and blue etc



(Apple also lost trying to fight RED)

RED/Nikon/Apple are still mates though as this is actually how the ProRes RAW formats work today, *actual need* also **becomes redundant over time**

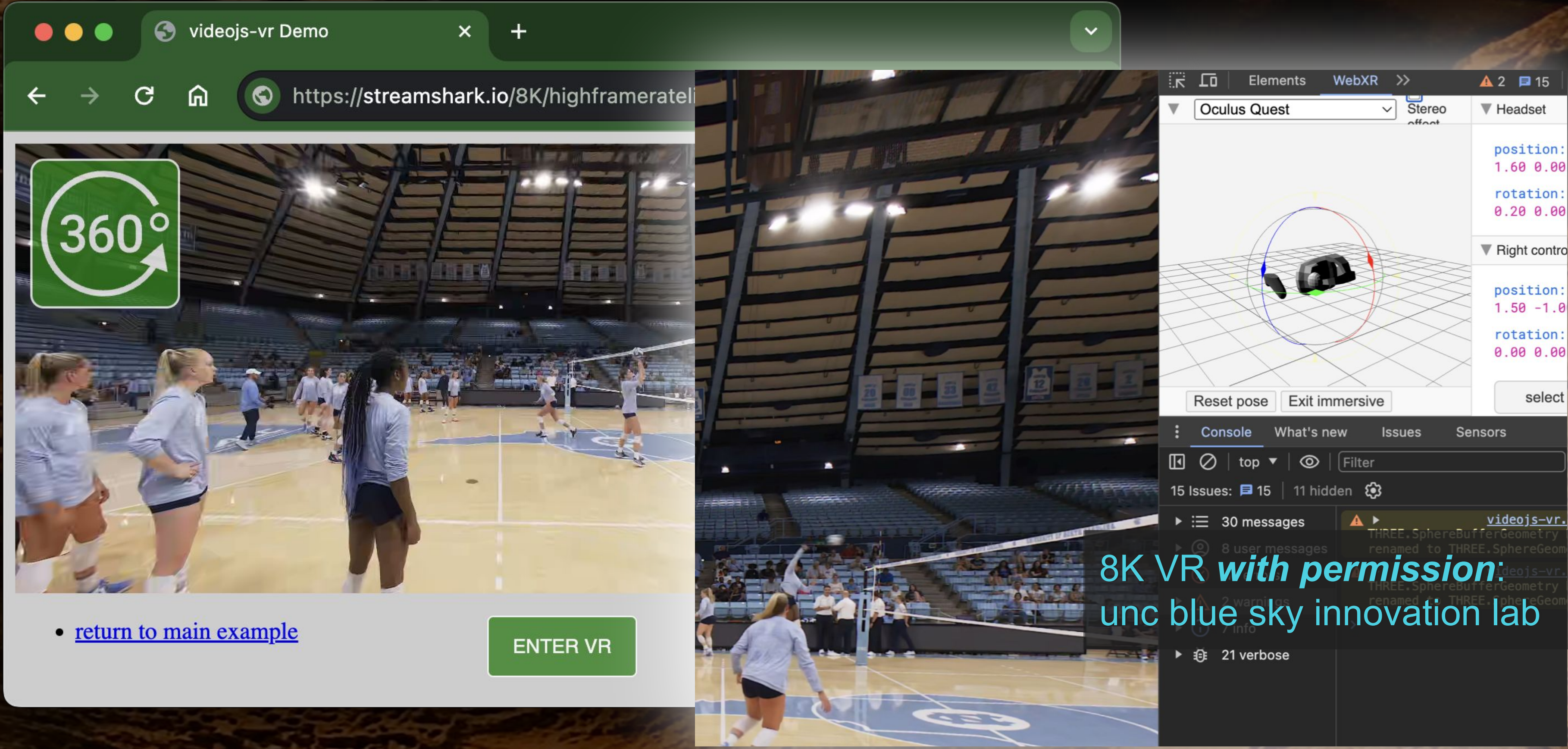
8K minefield, so it's all just doom and gloom?!

Geez Kev you're really selling 8K to us here :-)

Why bother?! go have beer instead - sit down mate, calm your farm and wait for it all to blow over...

... so what's Kev's ulterior motive

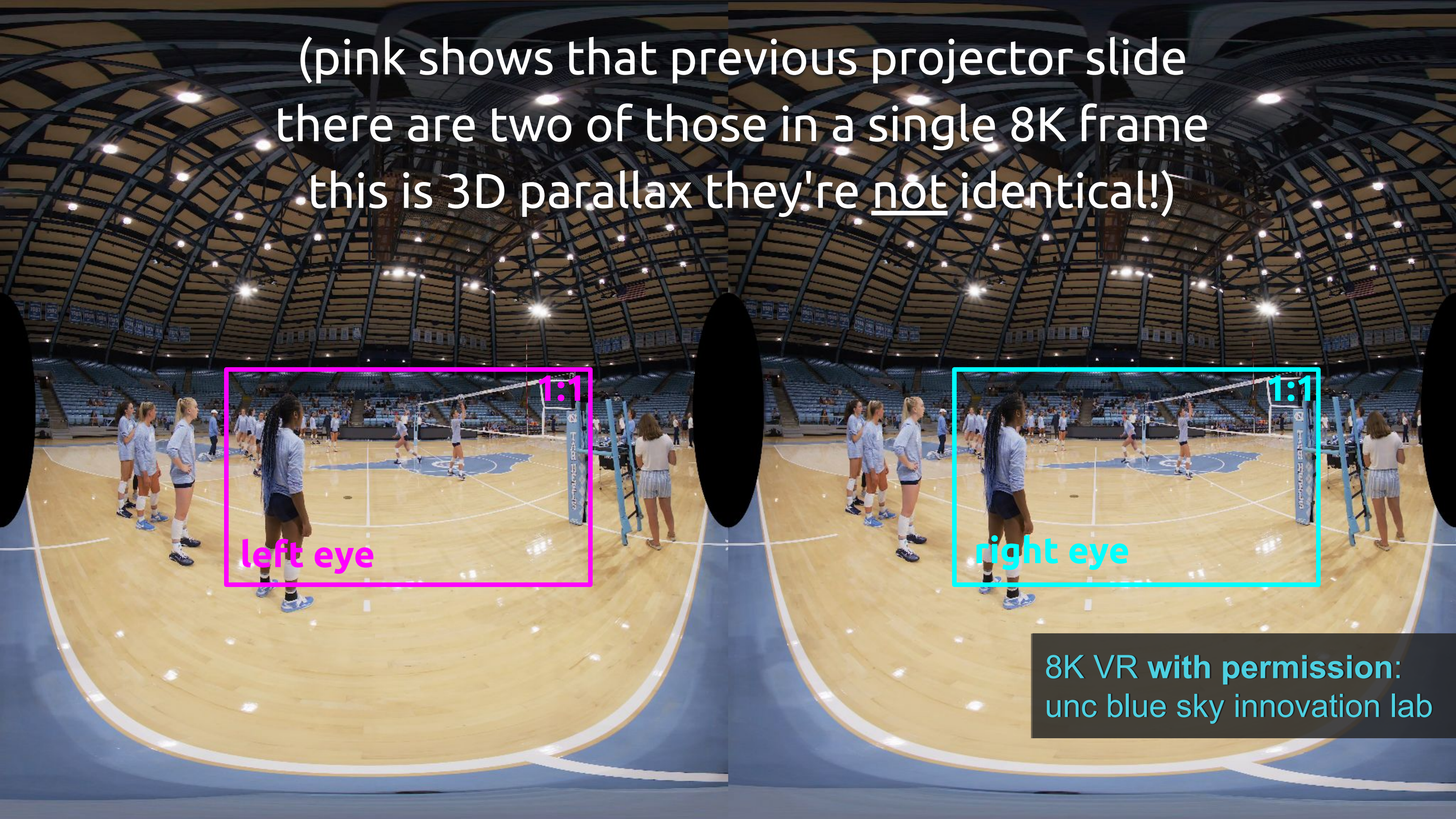
Epic VR! Enjoyable 4D immersive experiences



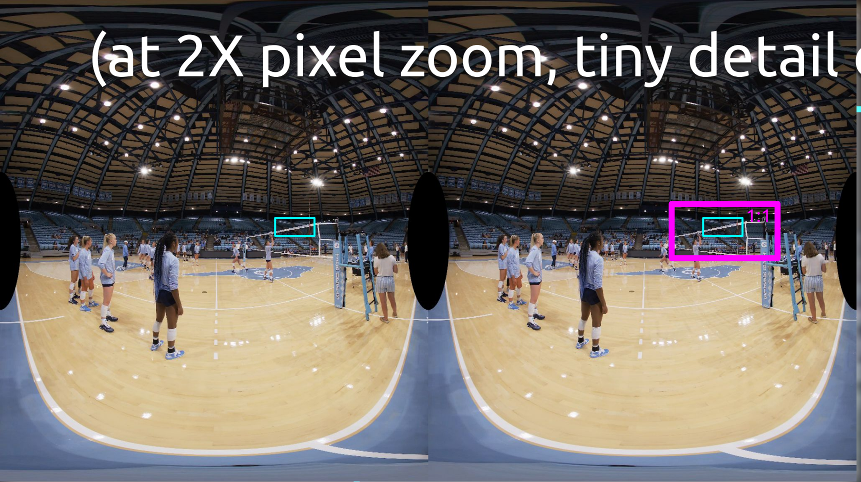
8K VR ***with permission***:
unc blue sky innovation lab

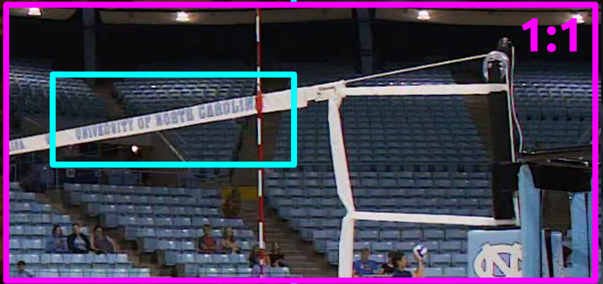

(stereo frame at **1/16th original size** for this 1080HD projector)
8K VR **with permission**:
unc blue sky innovation lab


1:1
(1:1 resolution, VR180° *without* any lens correction)
(*notice this is only* ***one*** *of the two eyes in that dual* square aspect cropped to 1920x1080
8K VR **with permission**:
unc blue sky innovation lab
TAR HEELS
ACC


(pink shows that previous projector slide
there are two of those in a single 8K frame
this is 3D parallax they're not identical!)
1:1
left eye
1:1
right eye
8K VR **with permission**:
unc blue sky innovation lab

(at 2X pixel zoom, tiny detail clarity also improves with movement)





8K VR **with permission**:
unc blue sky innovation lab

```
ffmpeg -i 8k60_stereo_180_volleyball.mp4 -vf "crop=250:250:2570:1700,scale=5000x5000" kevs_super_zoom.png
```

oh, a whole secondary camera rig too! (the other attack line)

8K VR **with permission**:
unc blue sky innovation lab

Volleyball court-side VR configuration

Shot 60fps with **two** RED cameras - setup on both court attack lines with stereoscopic lens

Stereo VR180° (3840px square aspect)

Looks absolutely **stunning** on VR headsets!

Bandwidth expectation ~ 50 Mbps (AV1) ~ 55 Mbps (HEVC) - no camera operators are needed on court

What if you don't have a spare $**200**K for the gear

Second hand gear? **good luck**! - not many 8K sensors around today... this is likely *not* going to be a redundant technology anytime soon

... but 4K cameras are becoming easier and cheaper to obtain - taping two together and strapping on fish eye lenses isn't so horrible

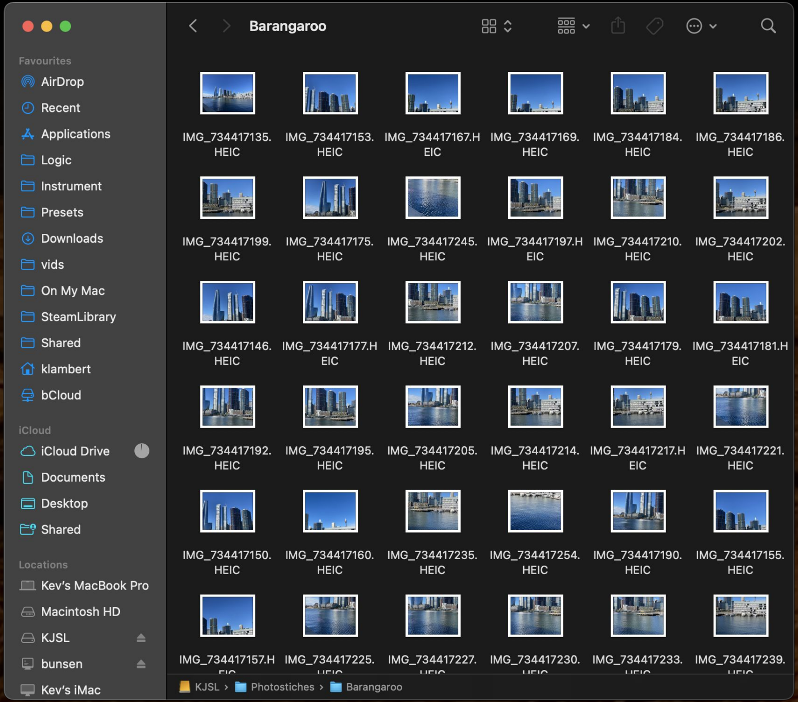
Roll your own 8K rectilinear - use 'phone camera!

Instead of taking one snap, take at least 4 extra shots around your subject ***without*** moving position

Stitch those as one with an app (there are several on app store)

A plumb bob is great for maintaining a fixed point in space

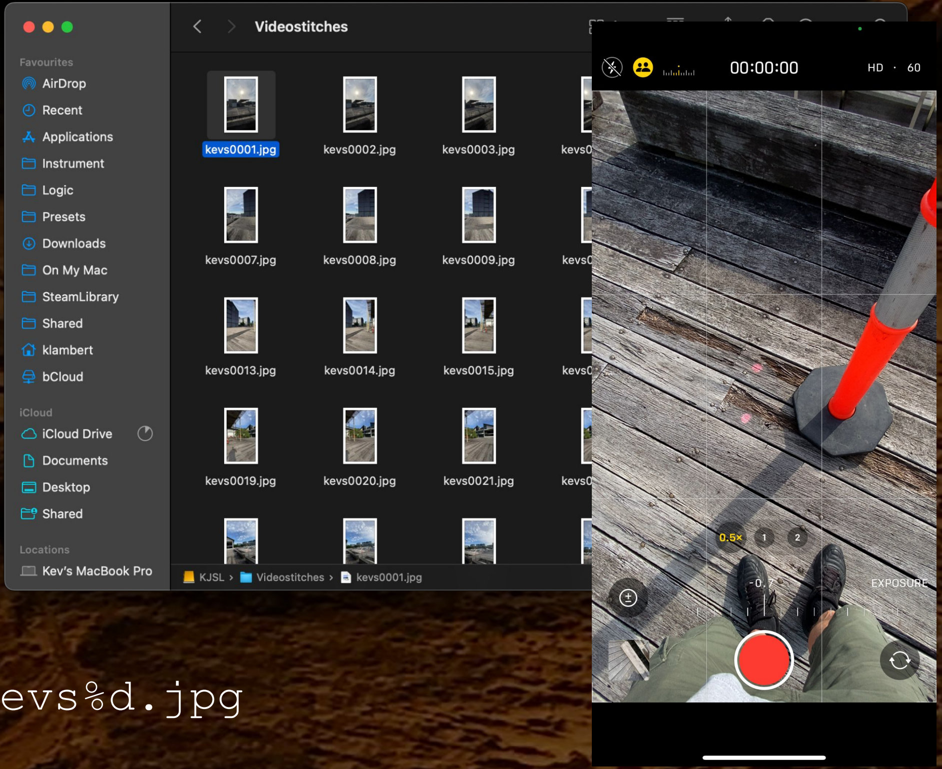
Don't forget to set your camera **exposure** to **locked!**



Roll your own 8K equirectangular - 'phone video!

Rather than taking snaps you can take a video walking around instead and auto generate snaps from that
You **must** find something to rest hand on e.g. a bollard to act as a gimbal/tripod

FFmpeg can create jpeg files from 4K video which you can import into stitching software such as PTGui




```
ffmpeg -i wharf.mov -vf fps=4 kevs%d.jpg
```

Roll your own 8K - Advanced stitching (PTGui)

(PTGui stitched preview)

Airdrop/copy to a PC - no filters
Convert RAW to JPEG -
preserving the EXIF metadata

magick mogrify -quality 100
-monitor -format jpeg *.HEIC

Project as equirectangular 180°
or 360° x 180° for full VR
Export as 3840x3840 square

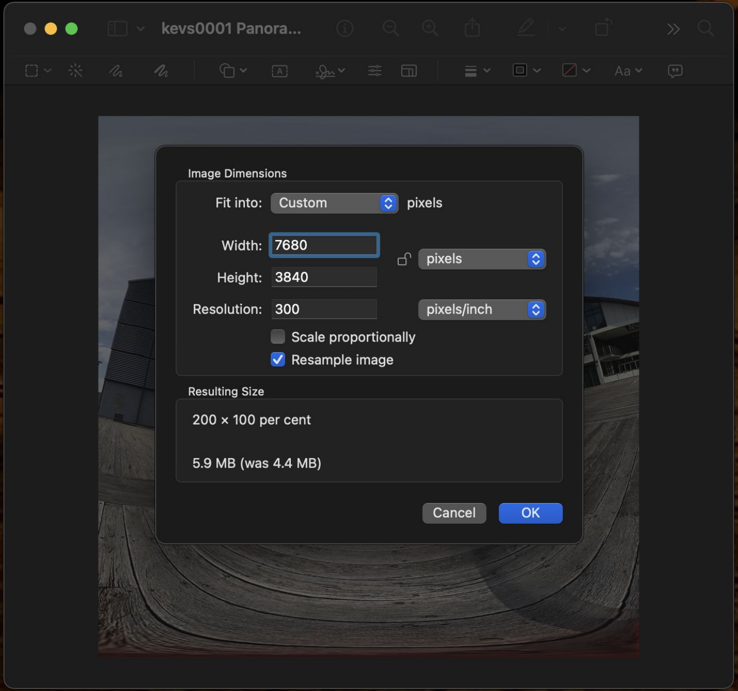
Roll your own 8K - Make Mono, Stereoscopic

If you are really (really) steady you can shoot a separate side-by-side for 3D stereoscopic

If not a depthless 8K VR180 still looks pretty good on headsets

Resize to 7680x3840 and duplicate (copy/paste)

I just used Preview here but can easily script using FFmpeg/Photoshop/Gimp



Roll your own 8K - et voilà stereoscopic VR spatial

Woah holup! that's an 8K *image* - not 8K *video!?*

Yep, ok you noticed... so let's use our static footage and make it **temporal**

Non-VR 8K - rectilinear simple crop animations are possible (Ken Burns pan/zoom)

VR equirectangular - dissolves and adding good ambient sound from each location **is still compelling**

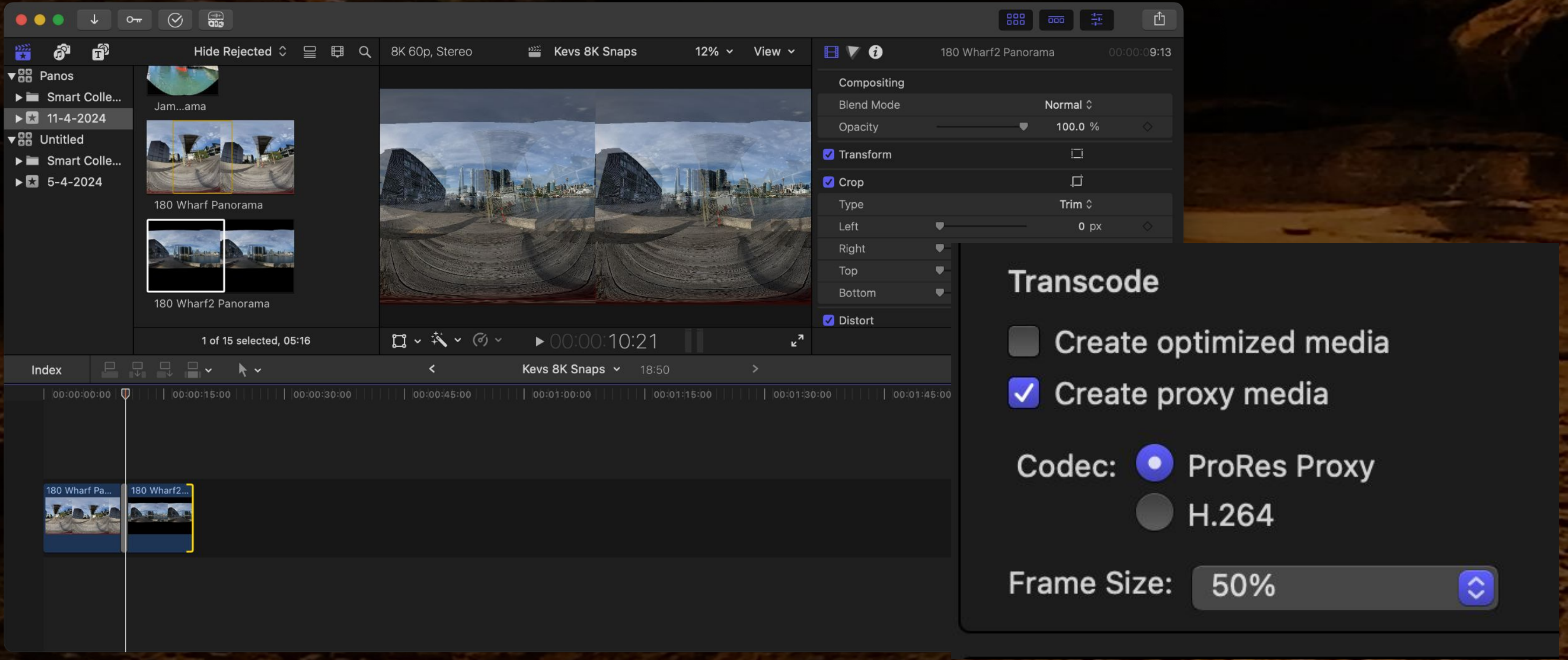
Roll your own 8K video - FCPX project setup

Roll your own 8K - FCPX 8K quality options

LT = Limited Storage
saves the most space

XQ = eXtreme Quality
video rendering (HDR)
aka another 4TB SSD

Roll your own 8K - FCPX don't forget to proxy!



What about *actual* 8K video capture on a budget

Possible cheat is using **machine learning** to colour-by-numbers those missing pixels for you

Various ways to do this, e.g. hardware but also free open source methods you can run locally for free

(watch out for hidden in/egress costs on top if choosing commercial cloud offerings)

180° Fisheye lens for 'phone camera (monoscopic)

Having tried this, not great results, VR 'tunnel' effects, but you might get luckier

'Phones often have multiple cameras too so choose the right one! Any EXIF data will be terribly wrong!

4K AI upscale - 8K Big Buck Bunny example

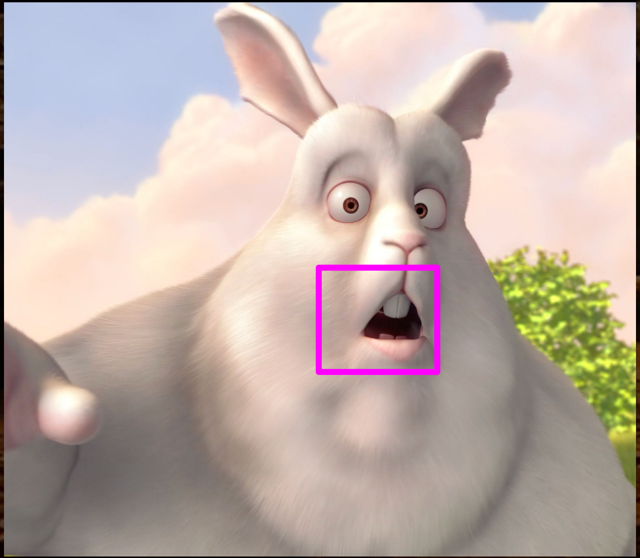
'Super-Resolution' algorithmically upscaled with ML

so, is *this* still 8K?

(pub discussion!)

#bbbcmif

8K AI upscale



Roll your own 8K - AI upscayl with ESRGAN

Sources: https://github.com/kevleyski/**upscayl**

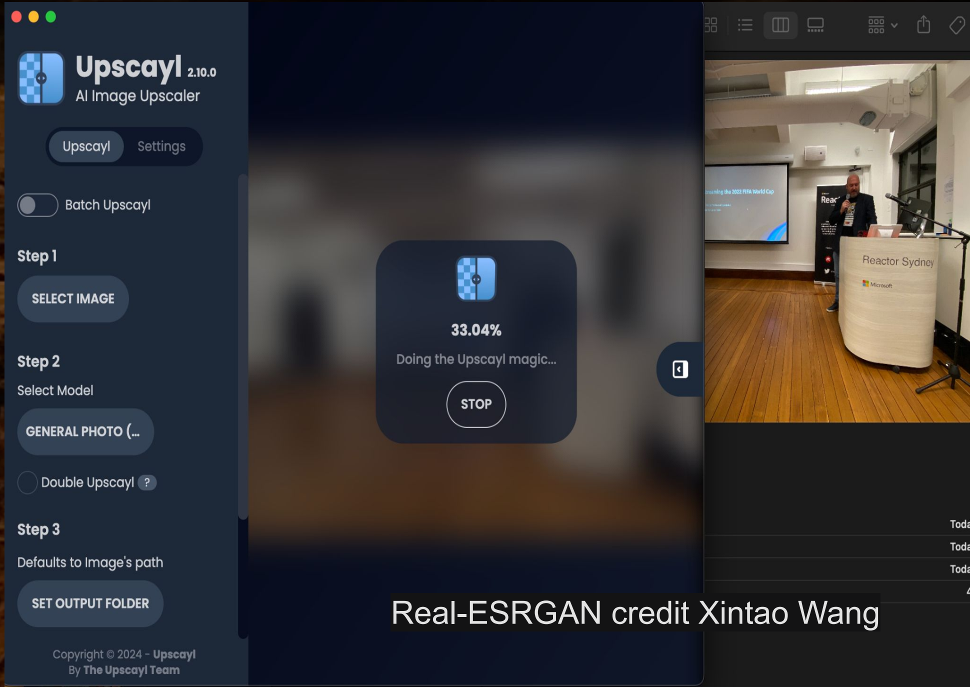
(node.js) `npm run start`

ESRGAN compares against original over and over providing feedback

Use FFmpeg extracts 4K to frames Super-Resolution output frames

BBB @ 60fps is many hours! :-)

Then hw encode the super res frames as 8K AV1/HEVC



Real-ESRGAN credit Xintao Wang


colin.png
"Electron" Notifications
Notifications may include alerts, sounds and icon badges.
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Document type: PNG image
File size: 12.7 MB (12,702,360 bytes)
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Modification date: 17 Apr 2024 at 12:10 PM
Image size: 4032 × 3024 pixels
Image DPI: 72 pixels/inch
Colour model: RGB
ColorSync profile: Display P3
1:1
Microsoft
SYD
Find us
Reactor Sydney
Microsoft
Original (cropped 1080HD)
4032x3024

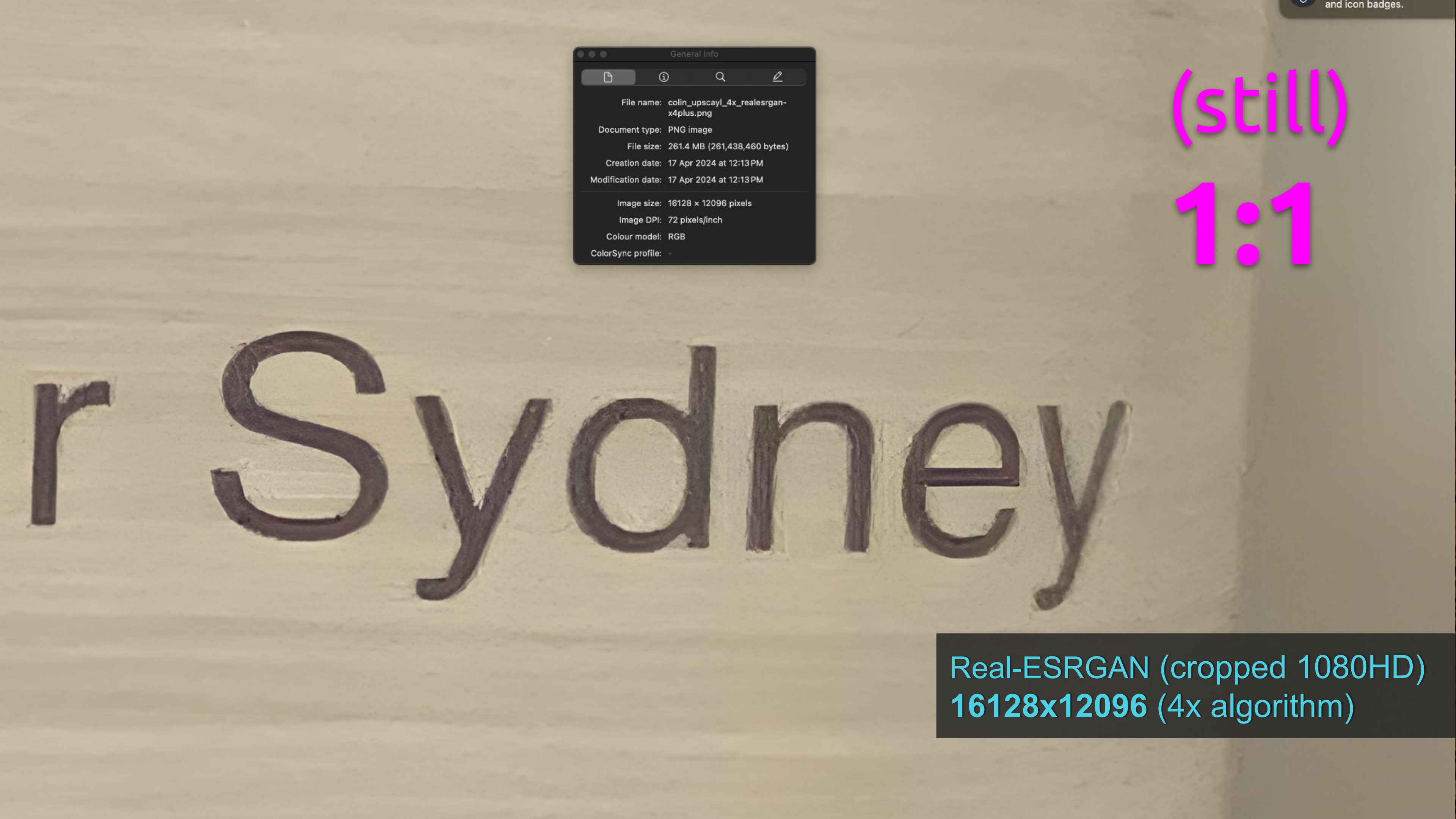
and icon badges.
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Image DPI: 72 pixels/inch
Colour model: RGB
ColorSync profile: -
(still)
1:1
r Sydney
Real-ESRGAN (cropped 1080HD)
16128x12096 (4x algorithm)

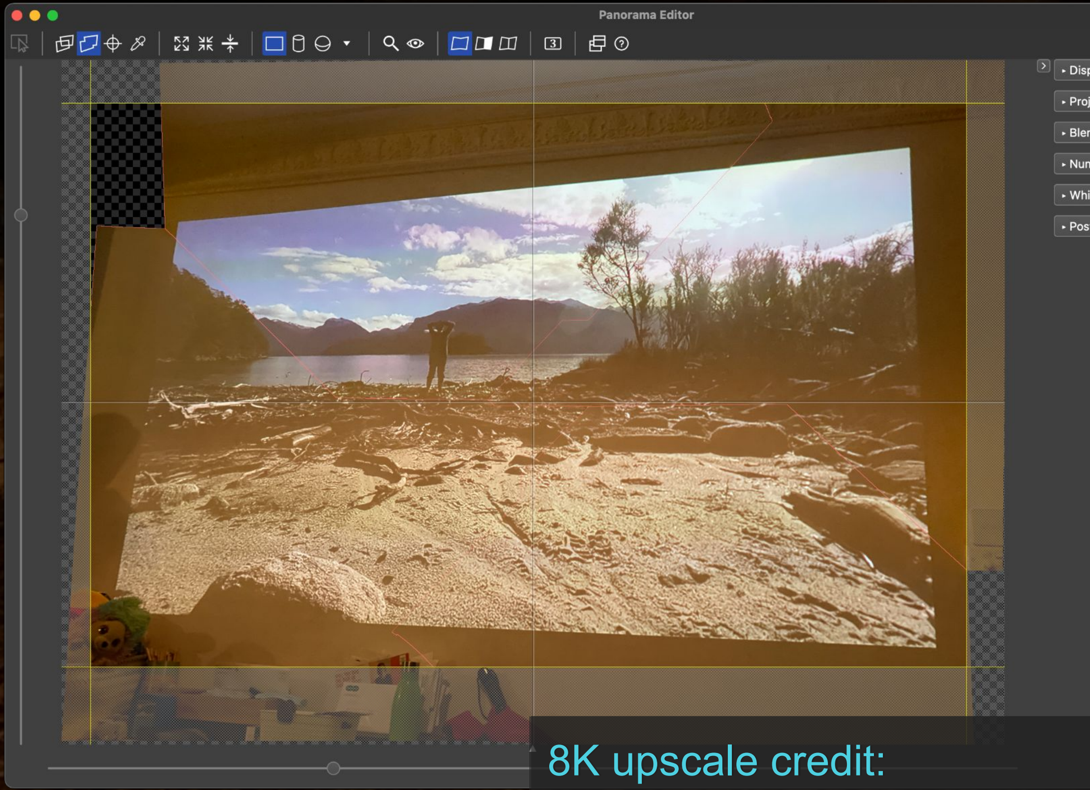
8K optical upscaling - 8K SBS "Alone Australia"?

8K photo taken of a 1080HD projection

so, is *this* still 8K?

(pub discussion!)

... just joshing, of course but it's technically 8K ;-)



8K upscale credit:
Kev (fairuse, I recon)

Roll your own 8K - AI text to 8K video

Pushing what is "video capture" a fair bit *but*...
an interesting idea is to generate 8K video from text prompts (AI text-to-video)

Meta AI - makeavideo.studio
RunwayML Gen 2 (Canva)

"Man with CC and 8K on tshirt"
(not 8K output or fisheye but hey, it could be!)

Roll your own 8K - live **8K** eSports

- High quality 8K eSports also possible via "headless WINE" on the encoder to generate large arena style spectator content from (**unmodified**) well known existing Windows/Intel games like Counter Strike 2
- Multiple arenas with different scenarios and "camera" angles would be compelling content

Roll your own 8K - live **8K** "headless WINE"

OpenSource WINE **wined3d** translates
Win32 API (*no licence is needed*) to POSIX API
OpenGL/Vulkan/Metal

OS/architecture agnostic
WINE runs fine on
Intel x86/AMD64
ARM/Apple Silicon M1/2/3
Linux/macOS
Android/iOS/xBSD etc

Windows DirectX12 on **macOS** and **Linux** (for **free**)

Steam (Windows) on Apple M3 Max (ARM)

WineSkins open source and CodeWeavers efforts, make using WINE easy

8K SBS "Alone Skyrim"

(multiple contestants *could* be 8K live rendered on **Linux** based encoders)

EULA:
https://bethesda.net/data/eula/en.html

Roll your own 8K idents, bumpers, interstitials

WebGPU has **very fast** side-by-side 4K blitting

HTML5/CSS3 (real smooth stereo transitions)

Low cost - 3D quality idents, bumpers and interstitials!

Things like Google Maps street view also have great definition VR (**monoscopic**)

Roll your own 8K idents, bumpers, interstitials

Be mindful of large bitrate ladder variance if inserting ads alongside your 8K content

Bitrates according to HLS spec are supposed to remain within 10% threshold, so your discontinuities should land around 46Mbps - 56Mbps too, as things scale bigger this detail gets forgotten about! *go talk to StreamShark*

Headless mode Browsers (8K on Linux encoder)

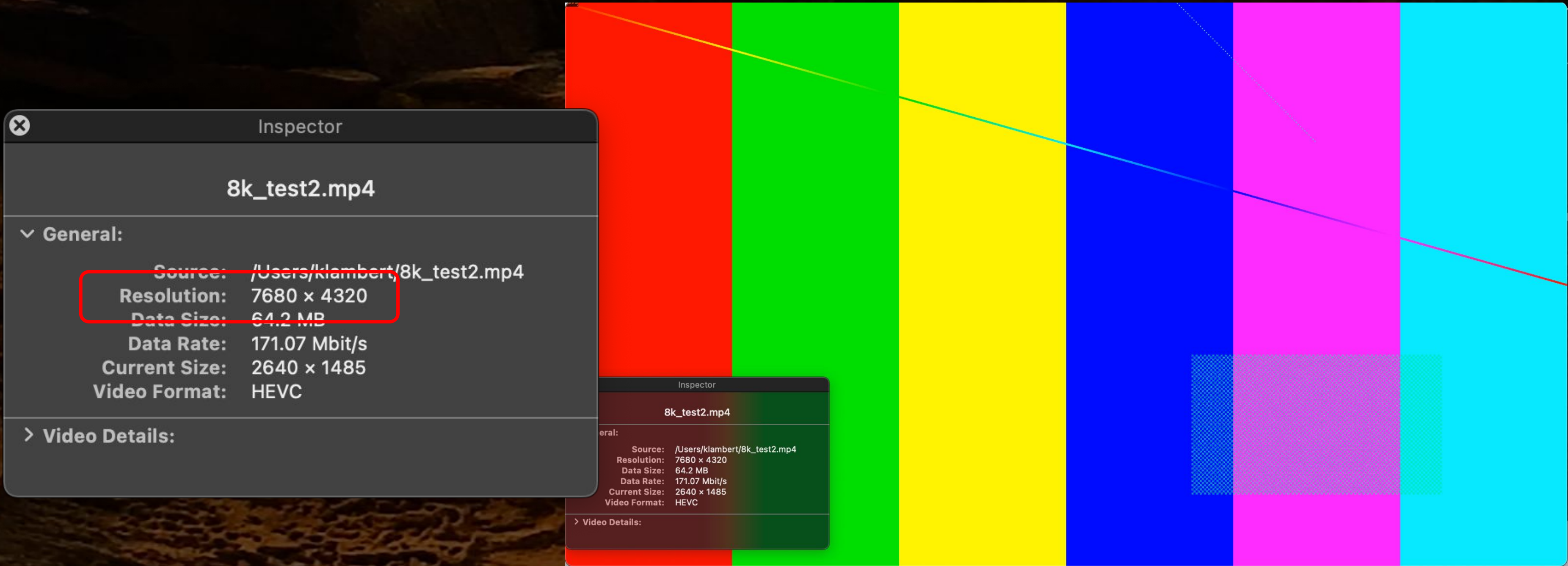
Unity supports fisheye distortions and WebGL

THREE.js and A-Frame can make projecting 2D onto spheres and amphitheatres easy

Run videojs-vr in **headless** in 8K res

8K FFmpeg synthetic input lavfi *(no VR projection)*

```
ffmpeg -y -f lavfi -i
testsrc2=duration=3:size=7680x4320:rate=60 -c:v
hevc_videotoolbox -tag:v hvc1 -b:v 13000K 8k_test2.mp4
```



8K needs hardware codecs, what are the options

- AV1 (AOMedia and SVT)
- H.265 HEVC (UHDTV)
- MV-HEVC
- VVC
- LCEVC (hybrid of hw and sw)

8K AV1

- NVIDIA Ada (2/3 x av1_nvenc split frame enc)
- AMD just announced at NAB 2024 new chips
- Intel Xe 2 (SVT-AV1 2.0) Rocket Lake+
- Apple M3+M4 ARM VideoToolbox (encode + decode) unless you're Netflix then well ok then it'll work on some older Apple devices too - or big bag of money maybe

8K AV1

Vindral just claimed world's first 8K AV1 live
We literally showed this 3 years earlier with SVT-AV1!

.. grrr, calm your farms
Kev and Colleen

NABshow 2024

8K H.265 / HEVC (High Efficiency Video Coding)

- NVIDIA Turing based cards will merrily spit out 8K 10-bit HEVC Main10 for you
- RDNA 3 (AMD) - 8Kp60 (Main10 enc + decode)
- Apple HEVC VideoToolbox Intel and M2 ARM (encode + decode)

8K MV-HEVC (multiview extensions)

M2 hw, captures depth view information alongside the stereoscopic video encoding

Not actually a new concept (2015), others including myself thought about it long before, spatial redundancy between two eyes is very high, duplicates already compress well though

8K MPEG-5 **LCEVC** (WebAssembly hw/sw hybrid)

Side loading 8K from 4K base encoding

Allows sharing same 4K encoding for legacy headsets, with 8K enhancement derived from a common 4K **hardware** decoder, this is rendered via WebAssembly in the browser

(spatial advantages too, talk to StreamShark)

8K VVC (multi-layer aggregated encoding)

VVC HD @ 1 Mbps +

VVC 4K @ 9 Mbps +

VVC 8K @ 25 Mbps = 8K @ 35 Mbps

To view the 8K you (need to) combine all three layers as composite to get the final result. Splits out your encoding ladder nicely

8K Live Scheduler from StreamShark

Server-side seamless stitching of 8K clips!

Zero encode - 8K dailies/remote cutting room

Choose your own adventure - more engaging interactive targeted ads and personalisations

Integrate Adobe Firefly/Canva Magic edits into your StreamShark workflow

Conclusions - Codecs will get better

8K res is not really needed for the whole immersive image
As experiment put your arms out and work out when you see motion first - that's your fight / flight cognitive reaction great for jump scare apps but when you *actually* see it's your fingers, that's not the same thing

lots of redundancy with VR - maybe some future codecs might well compress better for this, watch this space! ;-)

Conclusions - Prices will fall

Inevitably price of 8K cameras
and storage to capture from the sensors will come down,
8K stereoscopic GoPro's? *(not today AFAIK, but pretty likely)*

Demand for 8K *might* never be all that high

- certainly if there is little content and no need for 8K TVs

Headsets also **don't** need to be 8K to render! they only need to decode 8K. Capture quality is what really matters

Conclusions - Quality 8K capture will create demand

So perhaps future proof your photos from today - if you think you have a great snap that folks in the future will think is pretty snazzy then take maybe 4 additional at same optical zoom around your main subject

8K *needs you*!

Parting thoughts - Watch out for parallax

Whilst a plumb bob **does help**, sometimes close foreground objects will be plain just too hard to capture right unless you have a super steady hand, get a tripod

To be honest stereoscopic is *near impossible* without the right gear, the challenge is on!

and *probably* a world's first... **Aqua Rugby in 8K!** :-)